XINYU HUANG

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EDUCATION

The Chinese University of Hong Kong (QS 18) Hong Kong SAR, China

Master of Science in Advanced Studies in Statistics and Data Science Sept. 2025 – Nov. 2026

Average Score: 3.725/4.0

Relevant Courses: Advanced Statistical Inference (PhD level, A), Statistical Machine Learning (A-), Optimization in Data Science (A-), Bayesian Statistics with Applications in Machine Learning(A-), Applied Linear Models for Data Science(A-)

Northeastern University (Project 985) China

Bachelor of Science in Applied Statistics Sept. 2021 – Jul. 2025

Average Score: 89.087 / 100.000 (Ranking 4/20 in Class and Overall Ranking 6/41 in Department)

Relevant Courses: Biostatistics (99), Time Series Analysis (98), Probability Theory (97), Bayes Statistics (96), Statistical Practices (95), Statistic Computation Methods and Software (92)

Scholarships and Honors: Meiye Foundation Scholarship (11 out of 3100); Outstanding Graduate of The University; Merit Student of The University; Innovation and Entrepreneurship Scholarship; Second Prize Scholarship

PROJECT EXPERIENCE

A SQL-Based Hybrid Database-Driven Framework for Customer Satisfaction Prediction with Machine Learning and Natural Language Processing -Enhanced Sentiment Analysis **Sept. 2025 - Nov. 2025**

- Using SQL, reverse engineering and reengineering were conducted on 8 core tables of the e-commerce database. An ER diagram was designed and the data of orders, payments, logistics and comments were integrated to build a unified analysis view. Missing value processing, abnormal order elimination and feature derivation based on business logic (such as freight efficiency ratio and product volume) were completed to construct a high-quality modeling dataset.
- A prediction pipeline was built and three models, Ridge Regression, RF and XGBoost, were compared. Feature importance analysis and screening were carried out through SHAP values and mutual information method, and the Random Forest model performed the best, accurately quantifying the nonlinear impact mechanism of logistics timeliness and payment amount on user ratings.
- A complete NLP process was implemented, including Portuguese text cleaning (noise removal, stem extraction), stop word filtering and TF-IDF vectorization. MultinomialNB, Logistic Regression and RF were compared, and Logistic Regression was finally selected for deployment because it maintained a high accuracy rate (89.28%) while having excellent model interpretability.
- Combining the RFM model and EDA analysis, the risks of low user retention rate, uneven regional development and excessive reliance on credit card payments on the platform were revealed. Data-driven operation strategies such as "delayed compensation for high-value orders", "control of freight ratio threshold" and "awakening of dormant users" were proposed to guide business to optimize user experience.

RESEARCH EXPERIENCE

A Theoretical Understanding of Vision Transformer from A Foundation Model Perspective Apr. 2026

- Conducted a theoretical analysis of shallow single-head and multi-head Vision Transformers (ViTs) based on binary classification models. For single-head ViT, revealed that the sample complexity required to achieve zero generalization error is inversely proportional to the proportion of the square of relevant labels, and positively correlated with label noise level and initialization error. For multi-head ViT, increasing the number of attention heads reduces the sample complexity needed for zero generalization error, particularly in scenarios with sparse label information or high noise levels, and selecting the number of heads that matches the target task leads to better performance. Moreover, appropriately removing irrelevant and noisy labels can enhance model performance to a certain extent.
- In simulation experiments, validated the four conclusions for single-head ViT. Additionally, when the proportion of relevant labels decreases, the sample complexity for multi-head ViT grows exponentially

A Comparison Between Gray Models of The Conformable and Caputo Types Using Shock Data May 2025

- Introduced the Conformable fractional-order accumulation operator, derived its simplified coefficient form and time response, and established the CoFGM(r,1) model; validated improved fitting accuracy and trend prediction in real-case studies
- Introduced the Caputo-based fractional-order accumulation operator, revealing the relationship between order and new-information priority; derived the time response via Laplace transform to construct the CaFGM(r,1) model, achieving better global information integration and initial-value influence characterization
- Designed a multi-scale oscillatory test dataset and conducted systematic comparisons among GM(1,1), CoFGM(r,1), and CaFGM(r,1); results show CoFGM(r,1) achieves optimal sensitivity–efficiency trade-off and performs best under non-stationary and oscillation scenarios

A Functional Data Classification Model Utilizing Functional Mahalanobis Distance and Regenerative Kernel Methods Oct. 2023

- Developed a similarity measurement method for functional data based on functional Mahalanobis distance and regenerative kernel theory and applied it to functional kernel principal component analysis
- Discussed the application of this similarity measurement in other machine learning algorithms based on regenerative kernel theory and the development of corresponding analysis methods for functional data
- Combined Random Forest to classify functional data and compared its results with Euclidean distance-based reproducing kernel classification and B-spline Euclidean distance-based reproducing kernel classification, achieving an accuracy of 90% with a variance of 0.00096

COMPETITION EXPERIENCE

2024 Mathematical Contest in Modeling
Award: Meritorious Winner May 2024

- Explored the impact of lamprey sex ratio changes on the ecosystem using an ABM model based on Logistics-Verhulst equations
- Analysed the strengths and weaknesses of the lamprey population through an EWM model
- Constructed a food web ecosystem model to identify five indicators affecting ecosystem stability
- Simulated changes in sex ratios using differential equations based on the Lotka-Volterra multi-species host-parasite model, identifying stable solutions and demonstrating the role of sex ratio changes in maintaining food chain balance and resource stability
- Validated model simulations using biological data from Hammond Bay Biological Station's artificial streams, explaining the robustness of the model to various parameters and summarising its advantages and limitations

2023 Asia and Pacific Mathematical Contest in Modeling
Award: Second Prize Jan. 2024

- Predicted the future development and analysed the influencing factors of new energy electric vehicles using grey relational analysis, the TOPSIS model based on entropy weights, ARIMA and partial least squares regression models
- Investigated and discussed the promotion of new energy electric vehicles and the contributions of various countries to the new energy industry

2023 Mathematical Contest in Modeling
Award: Honorable Mention May 2023

- Preprocessed the raw data and applied ARIMA and the Grey Prediction Model GM(1,1) to forecast the total number of Wordle users in the future, and used GBDT combined with a Support Vector Machine to predict the number of attempts for the word 'EERIE'
- Performed Pearson correlation analysis, which revealed that the number of users in hard mode is related to word frequency and syllable count, as well as clustered word difficulty using K-Means, concluding that common words are easier, while words with more repeated letters are more difficult

EXTRACURRICULAR EXPERIENCE

Northeastern University
Position: Student Assistant to the President Dec. 2023 – Dec. 2024

- Liaised with various departments to investigate student-raised issues, seek solutions, submit recommendations to university leadership, and communicate institutional responses back to students
- In response to student complaints about campus network connectivity, conducted on-site visits and surveys, sought support from the university's network center, and facilitated the upgrade and iterative improvement of the campus network, ultimately achieving 100% dormitory coverage

Mathematics and Statistics College Student Union
Position: President Sept. 2022 – Sept. 2023

- Managed 11 departments of the student union
- Planned and conducted the 6th Student Representative Assembly of the Mathematics and Statistics College and the 12th Student Union Handover Conference, with 450 attendees
- Organised various large-scale university events, including the preliminary and final rounds of the Northeastern University at Qinhuangdao Campus History and Knowledge Competition, involving 9 colleges and 850 participants

SKILLS

Programming Skills: Proficient in C++, Java in both Linux and Windows OS
Data Analysis Skills: Proficient in Python, SPSS, SQL, VBA in Excel
English Skills: IELTS (Overall Band Score 6.5; Listening 6.5; Reading 7; Writing 6; Speaking 6), CET-6
Office software: Proficient in using Excel (Data Pivot, Vlookup), PowerPoint, Word, Lark, DingTalk, etc. Can write using LATEX.